College Station, TX
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Koya Yamamoto

SUMMARY

PhD candidate in Aerospace Engineering with a strong foundation in control theory, state estimation, trajectory optimization, and hardware-in-the-loop validation. Experienced in developing and validating advanced guidance, navigation, and control algorithms — from rigorous mathematical modeling to flight-ready, embedded software.

EDUCATION

Ph.D., Aerospace Engineering 2022 – Present

Texas A&M University, College Station, TX · GPA 4.0/4.0

- Selected coursework: Random Dynamical Systems, Space Domain Awareness, Celestial Mechanics, Spacecraft Dynamics, Optimal Spacecraft Attitude & Orbital Maneuvers, Estimation of Dynamic Systems.

B.S., Mechanical Engineering 2018 – 2022

Waseda University, Tokyo, Japan · GPA 3.6/4.0

- Selected coursework: Mechatronics, Fluid Dynamics, Satellite Design, C Programming.

TECHNICAL SKILLS

Programming	MATLAB, Simulink, Python, C++
Control & Estimation	Model Predictive Control (MPC), PID, Optimal Control, Kalman Filtering (EKF, UKF), Least-Squares Estimation
Optimization	IPOPT, SNOPT, fmincon, CasADi
Hardware	Arduino, Raspberry Pi
CAD & Analysis	SolidWorks, AutoCAD, Femap, Nastran
Other	LaTeX, Microsoft Office Suite

EXPERIENCE

Navigation Engineer Intern May – Aug 2023

VectorNav Technologies, Dallas, TX

- Developed smoothing algorithms for post-Kalman filtering, achieving a 2× improvement in state-estimation accuracy for real-time navigation systems.
- Created high-precision geoid-undulation algorithms using the EGM2008 model.
- Derived recursive formulas for Legendre polynomials.

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2022 – Present

Texas A&M University, College Station, TX

- Designed nonlinear and shrinking-horizon MPC for trajectory correction under uncertainty for cislunar orbit transfers; now developing tube-based MPC to incorporate uncertainty into optimal control.
- Developed the Complex-Step (CX) differentiation method for highly accurate, efficient derivatives of dynamic systems.
- Integrated graph-coloring techniques into trajectory optimization for a 2× speedup in real-time path planning.
- Validated control and trajectory-optimization techniques in hardware-in-the-loop testing with Arduino and Raspberry Pi.
- Conducted state estimation via linear least squares with covariance analysis from a minimum-accelerometer configuration.
- Developed a system-identification algorithm approximating complex multidimensional functions with analytical forms.

Technical Trainee Summer 2021

JAXA / ISAS, Kanagawa, Japan

- Modeled liquid-sloshing effects on spacecraft lunar-landing dynamics using physics-based mass-spring-damper models.

- Conducted simulation studies to determine system stability boundaries and failure modes.
- Proposed and validated control laws for landing-gear systems to prevent vehicle overturn.

Undergraduate Research Project 2020 – 2021

Waseda University, Tokyo, Japan

- Designed stratospheric-balloon experiments to validate an alignment-monitoring system.
- Conducted structural analysis for stratospheric conditions using Femap and Nastran, and performed CAD/SolidWorks design.
- Built and tested physical prototypes.

PUBLICATIONS

1. K. Yamamoto, E. Taheri, and J. L. Junkins. “Nonlinear MPC for Cislunar Trajectory Correction Under Impulsive Maneuver Uncertainties: A Computational Performance Analysis,” *American Control Conference (ACC)*, 2026 (to appear).
2. K. Yamamoto, E. Taheri, and J. L. Junkins. “Jacobian Coloring for Multi-Impulse Cislunar Trajectory Optimization,” *Journal of Aerospace Science and Technology*, Aug. 2025. doi:10.2139/ssrn.5545742
3. K. Yamamoto and J. L. Junkins. “Advances in High-Precision Partial Derivative Approximation for Black-Box Functions,” *Journal of Astronautical Sciences*, June 2025. doi:10.21203/rs.3.rs-6926796/v1
4. K. Yamamoto, E. Taheri, and J. L. Junkins. “Comparison of NLP Solvers and Derivative Accuracy for Solving Multi-Impulse Cislunar Trajectory Optimization Problems,” *2025 American Control Conference (ACC)*, IEEE, July 2025, pp. 3807–3812. doi:10.23919/ACC63710.2025.11107869
5. K. Yamamoto, P. Kelly, and J. L. Junkins. “Advancing Complex Differentiation: Applications to the State Transition Matrix,” *AAS/AIAA Astrodynamics Specialist Conference*, Broomfield, CO, Aug. 2024.
6. K. Yamamoto, P. Kelly, and M. Majji. “Covariance Analysis of Attitude Estimation Using Accelerometers,” *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, MT, Aug. 2023. arXiv:2502.00513

LANGUAGES

Japanese (Native) · English (Fluent) · Italian (Advanced) · Spanish (Beginner)